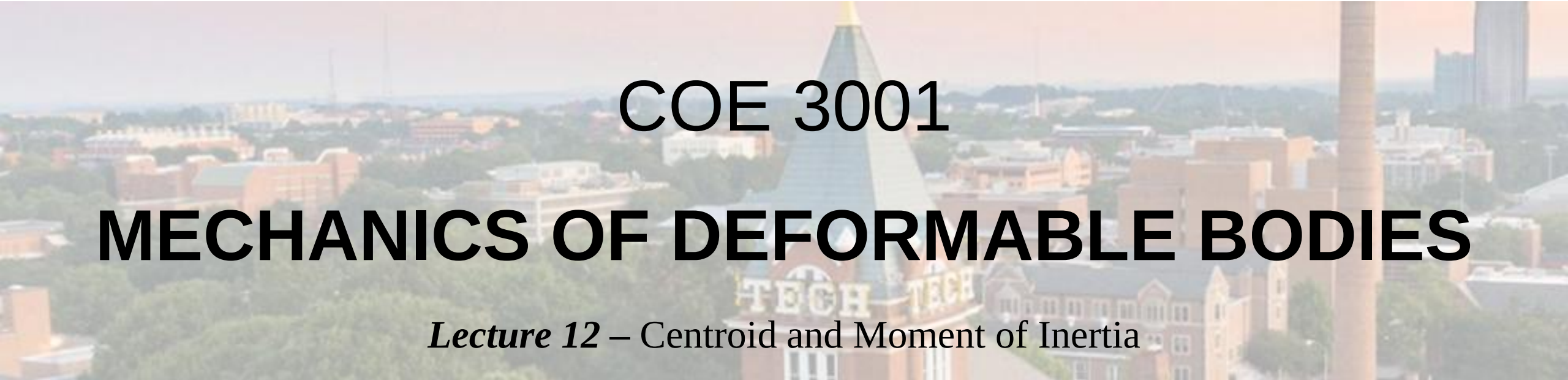




Georgia Tech
College of
Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus. The image shows various red-brick buildings, green trees, and a tall smokestack in the distance. The text is overlaid on this image.

COE 3001

MECHANICS OF DEFORMABLE BODIES

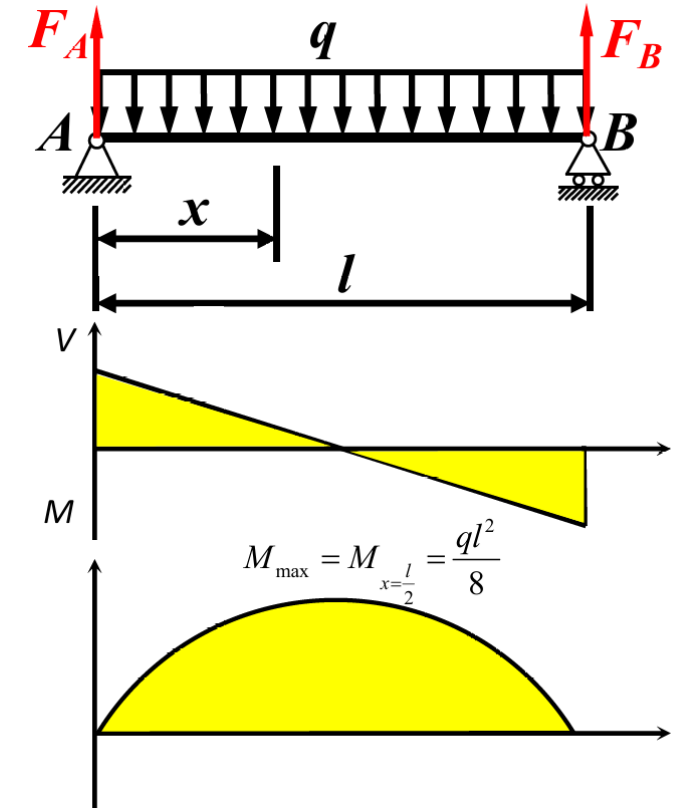
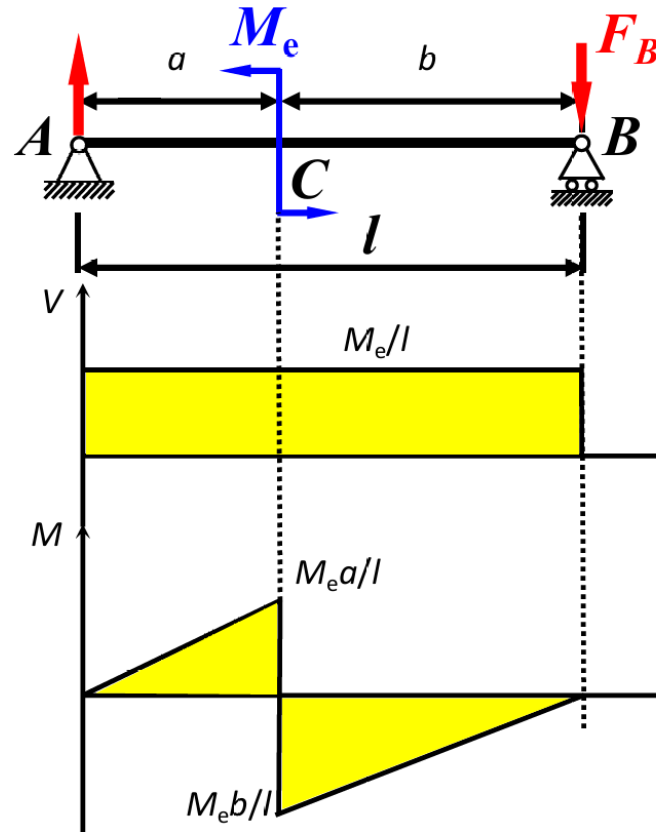
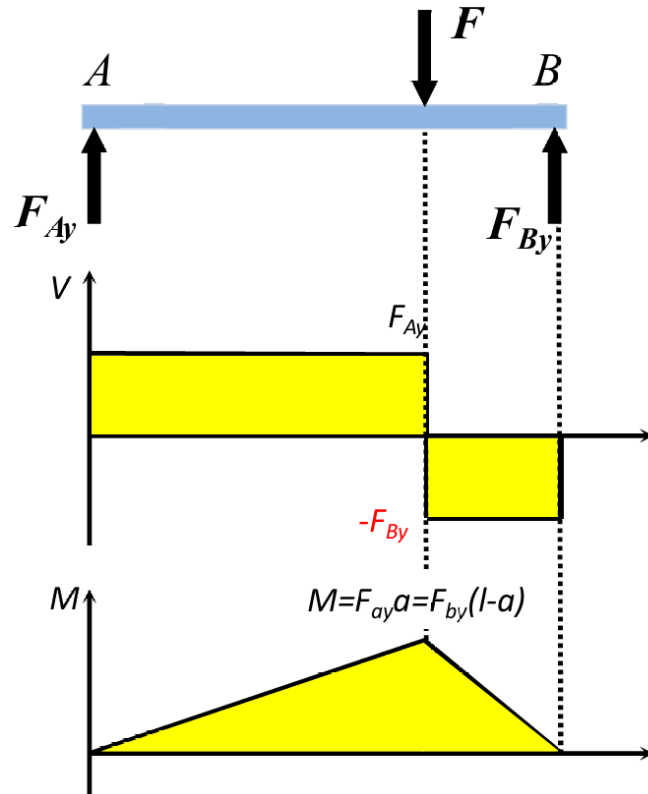
Lecture 12 – Centroid and Moment of Inertia

Yazhuo Liu

Georgia Institute of Technology

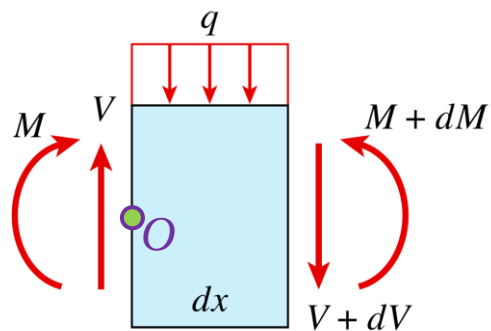
Mar. 2, 2026

Relationships among $q(x)$, $V(x)$, and $M(x)$



Relationships among $q(x)$, $V(x)$, and $M(x)$

$$\Sigma F_{\text{vert}} = 0 \quad V - q dx - (V + dV) = 0$$



$$\Rightarrow \frac{dV}{dx} = -q \Rightarrow V_B - V_A = -\int_A^B q dx$$

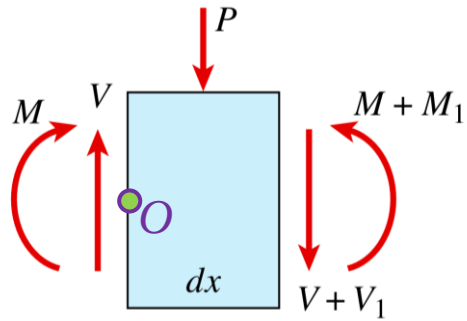
$= -(\text{area of the loading diagram between } A \text{ and } B)$

$$\Sigma M = 0 \quad -M - q dx \left(\frac{dx}{2} \right) - (V + dV) dx + M + dM = 0$$

$$\Rightarrow \frac{dM}{dx} = V \Rightarrow M_B - M_A = \int_A^B V dx$$

$= (\text{area of the shear-force diagram between } A \text{ and } B)$

Jumps due to the concentrated load



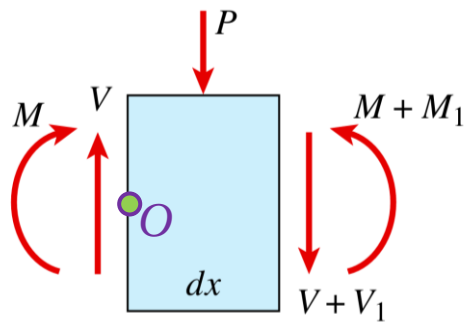
$$\Sigma F_{\text{vert}} = 0 \quad V - P - (V + V_1) = 0 \quad \Rightarrow \quad V_1 = -P$$

$$\Sigma M = 0 \quad -M - P\left(\frac{dx}{2}\right) - (V + V_1)dx + M + M_1 = 0$$

$$\Rightarrow \quad M_1 = P\left(\frac{dx}{2}\right) + Vdx + V_1dx$$

Since the length dx of the element is infinitesimally small, the increment M_1 in the bending moment is also infinitesimally small. Thus, **the bending moment does not change when passing through the point of application of a concentrated load.**

Jumps due to the concentrated load

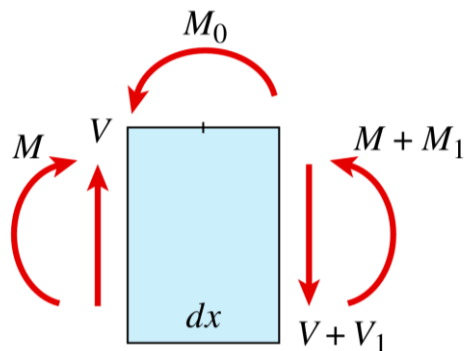


$$\Sigma F_{\text{vert}} = 0 \quad V - P - (V + V_1) = 0 \quad \Rightarrow \quad V_1 = -P$$

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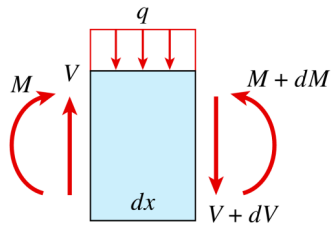
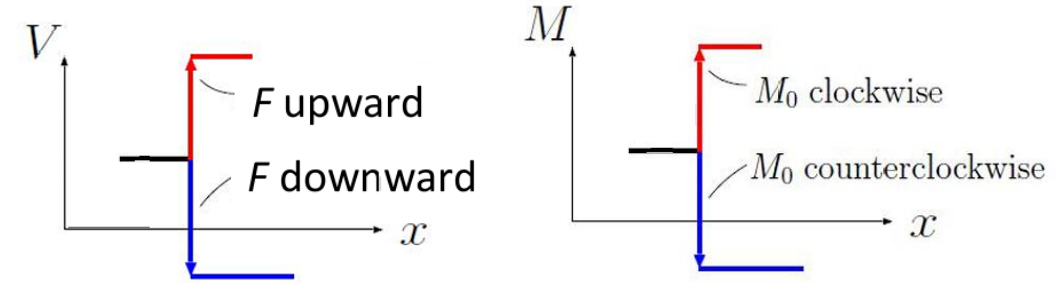
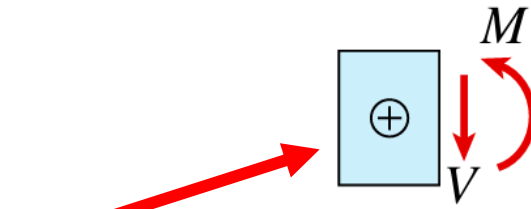
$$\Sigma M = 0 \quad -M + M_0 - (V + V_1)dx + M + M_1 = 0$$

$$\Rightarrow \quad M_1 = -M_0$$

This relation shows that the bending moment decreases by M_0 when moving from left to right through the point of load application. Thus, **the bending moment changes abruptly at the point of application of a couple/moment.**

Summary

From **left to right !!!!!**



$$\frac{dV}{dx} = -q(x) \quad q(x) \text{ downward}$$

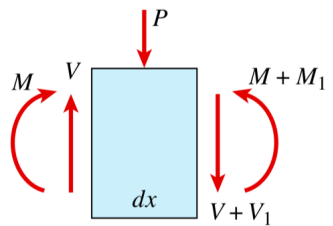


Upward $q(x) \rightarrow V(x)$ increase
Integrate $q(x)$ to get change in $V(x)$

$$\frac{dM}{dx} = V(x)$$



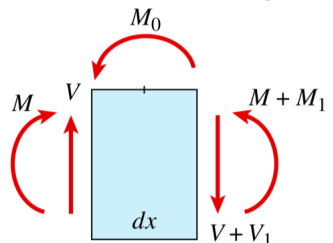
Integrate $V(x)$ to get change in $M(x)$



$$\Delta V = -P \quad P \text{ downward}$$



Upward $P \rightarrow V(x)$ upward jump



$$\Delta M = -M_0 \quad M_0 \text{ counterclockwise}$$

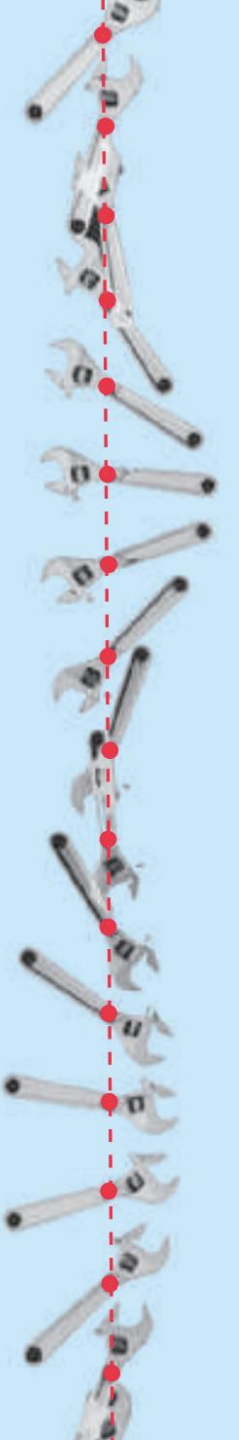


Clockwise $M_0 \rightarrow M(x)$ upward jump

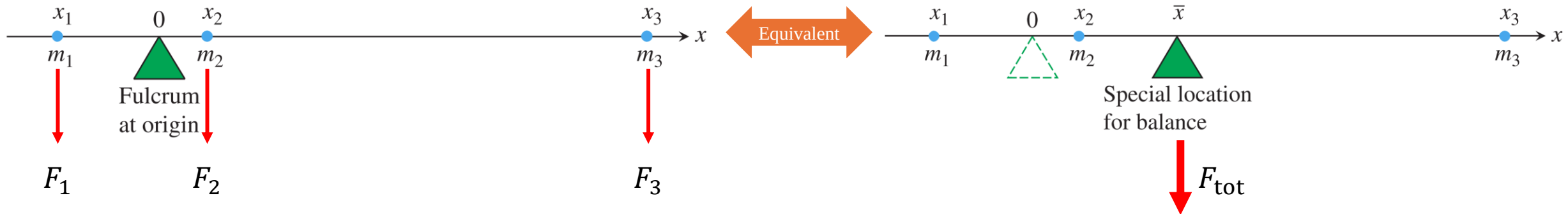
Centers of Mass

Many structures and mechanical systems behave as if their masses were concentrated at a single point, called the **center of mass**.

- It is important to know how to locate this point, and doing so is basically a mathematical enterprise



Masses along a line (Assume constant density and gravitational acceleration)



Total moment of the origin:

$$F_1 x_1 + F_2 x_2 + F_3 x_3$$



$$g \cdot (m_1 x_1 + m_2 x_2 + m_3 x_3)$$

Total moment of the origin:

$$F_{\text{tot}} \bar{x}$$



$$g \cdot (m_1 + m_2 + m_3) \bar{x}$$

$$g \cdot (m_1 x_1 + m_2 x_2 + m_3 x_3) = g \cdot (m_1 + m_2 + m_3) \bar{x}$$

Center of mass

$$\bar{x} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} = \frac{\sum m_k x_k}{\sum m_k} = \frac{\int x dm}{\int dm} = \frac{\rho \int_A x dA}{\rho \int_A dA} = \frac{\int_A x dA}{\int_A dA}$$

Distance to
y-axis

Static moment

Static moment (also called *first moment of area*) measures the spatial distribution of a shape in relation to an axis

$$S_y = \int_A x \, dA, \quad S_x = \int_A y \, dA$$

$$\bar{x} = \frac{\int_A x \, dA}{\int_A dA} = \frac{S_y}{A}, \quad \bar{y} = \frac{\int_A y \, dA}{\int_A dA} = \frac{S_x}{A}$$

For combination figures

$$S_y = \sum A_i \bar{x}_i, \quad S_x = \sum A_i \bar{y}_i$$

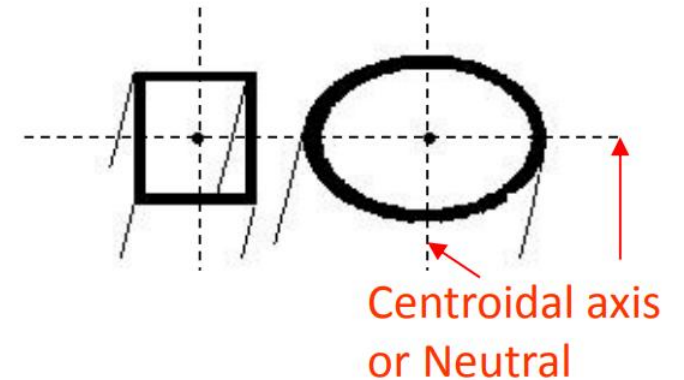
The subscript denotes which axis you're taking moment about.

When computing the moment, you use the perpendicular distance from the area element dA to that axis — which corresponds to the *coordinate along the other axis*.

Centroids

Centroid is the center of mass of a geometric object of uniform density.

- Usually coincides with center of gravity
- Centroid of 3D objects usually lies somewhere along the lines of symmetry
- Hollowed pipes, L-shaped section have centroid located outside of the material of the section



Centroids for composite figure

$$S_y = \sum A_i \bar{x}_i, \quad S_x = \sum A_i \bar{y}_i$$

$$S_y = \bar{x}_{\text{tot}} \sum A_i, \quad S_x = \bar{y}_{\text{tot}} \sum A_i$$

$\Downarrow$

$$\bar{x}_{\text{tot}} = \frac{\sum A_i \bar{x}_i}{\sum A_i}, \quad \bar{y}_{\text{tot}} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

3D
member

$$x_C = \frac{A_1 x_{1C} + A_2 x_{2C} + \cdots + A_n x_{nC}}{A_1 + A_2 + \cdots + A_n}$$

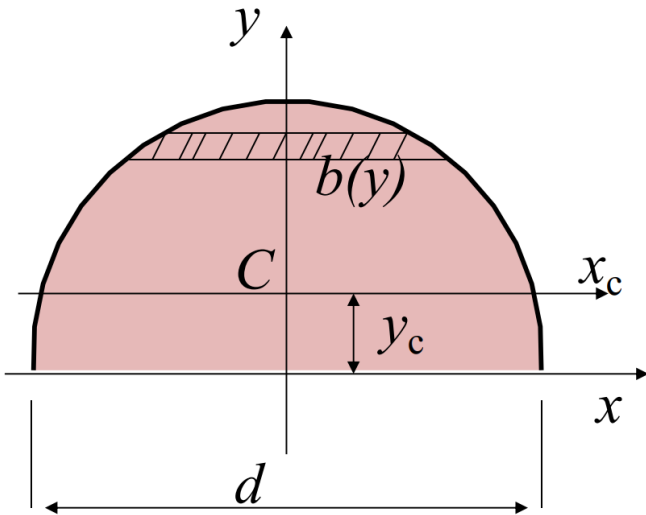
$$y_C = \frac{A_1 y_{1C} + A_2 y_{2C} + \cdots + A_n y_{nC}}{A_1 + A_2 + \cdots + A_n}$$

$$z_C = \frac{A_1 z_{1C} + A_2 z_{2C} + \cdots + A_n z_{nC}}{A_1 + A_2 + \cdots + A_n}$$

2D/plane
figure

Example

Determine the centroid of the semicircle



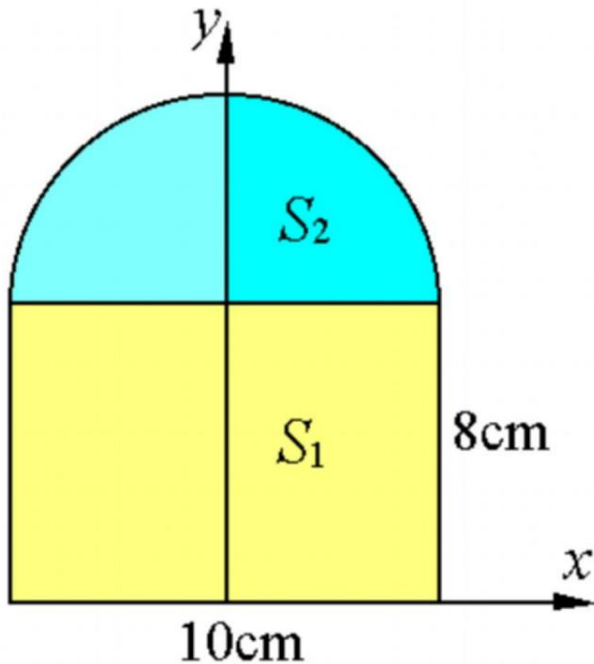
$$b(y) = 2\sqrt{R^2 - y^2}$$

$$\begin{aligned} S_x &= \int_A y dA = \int_0^{\frac{d}{2}} y b(y) dy \\ &= \int_0^{\frac{d}{2}} y \cdot 2\sqrt{R^2 - y^2} dy = \frac{d^3}{12} \end{aligned}$$

$$y_c = \frac{S_x}{A} = \frac{d^3/12}{\pi d^2/8} = \frac{2d}{3\pi}$$

Example

Determine the centroid of the composite figure



Solution:

1) set the symmetry axis as the coordinate axis, so $x_c=0$

2) divide the cross-section into two parts, S_1 and S_2

$$A_1=80\text{cm}^2, y_1=4\text{cm}$$

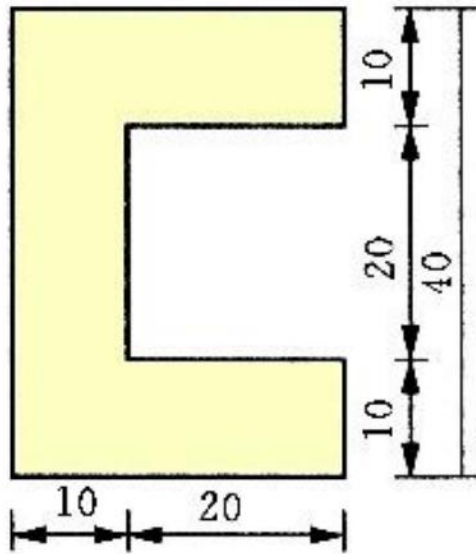
$$A_2 = \frac{1}{2}\pi R^2, \quad y_2 = \left(8 + \frac{4R}{3\pi}\right)\text{cm}$$

$$y_C = \frac{\sum A_i y_i}{A} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$\therefore y_C = 6.4\text{cm}$$

Example

Determine the centroid of the composite figure



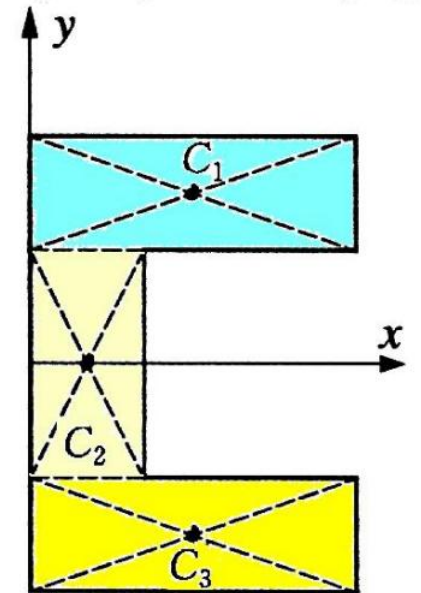
Solution: set the symmetry axis as the coordinate axis, so $y_c=0$. Then divide into three parts.

$$A_1 = 30 \cdot 10 = 300 \text{ cm}^2, \quad x_1 = 15 \text{ cm};$$

$$A_2 = 20 \cdot 10 = 200 \text{ cm}^2, \quad x_2 = 5 \text{ cm};$$

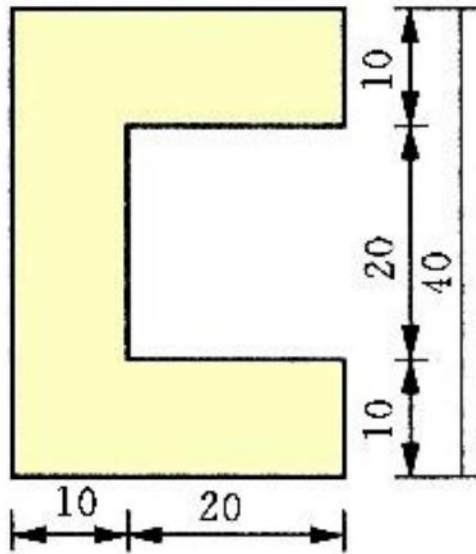
$$A_3 = 30 \cdot 10 = 300 \text{ cm}^2, \quad x_3 = 15 \text{ cm};$$

$$x_c = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} = 12.5 \text{ cm}$$



Example

Determine the centroid of the composite figure



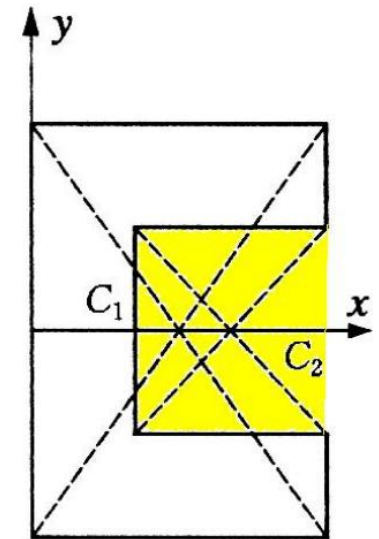
Solution: divide the cross-section into two rectangular shape, A_1 is positive area while A_2 is negative area.

Negative area method

$$A_1 = 30 \cdot 40 = 1200 \text{ cm}^2, \quad x_1 = 15 \text{ cm}$$

$$A_2 = -20 \cdot 20 = -400 \text{ cm}^2, \quad x_2 = 20 \text{ cm};$$

$$x_C = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = 12.5 \text{ cm}$$



Moment of inertia

Moment of inertia (also known as *the second moment of area*) is a term used to describe the capacity of a cross-section to resist bending.

- It is a mathematical property of a section concerned with a surface area and how that area is distributed about the reference axis.
- The reference axis is usually a centroidal axis

$$I_x = \int_A y^2 dA, \quad I_y = \int_A x^2 dA$$

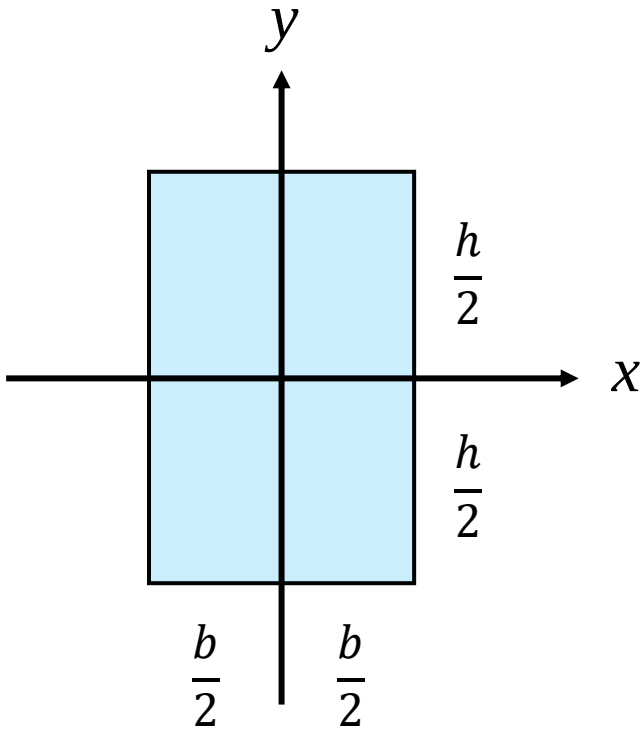
Recall:

$$I_p = \int_A \rho^2 dA = \int_A (x^2 + y^2) dA = I_x + I_y$$

Perpendicular axis theorem

Example

Determine the moment of inertia of the rectangle

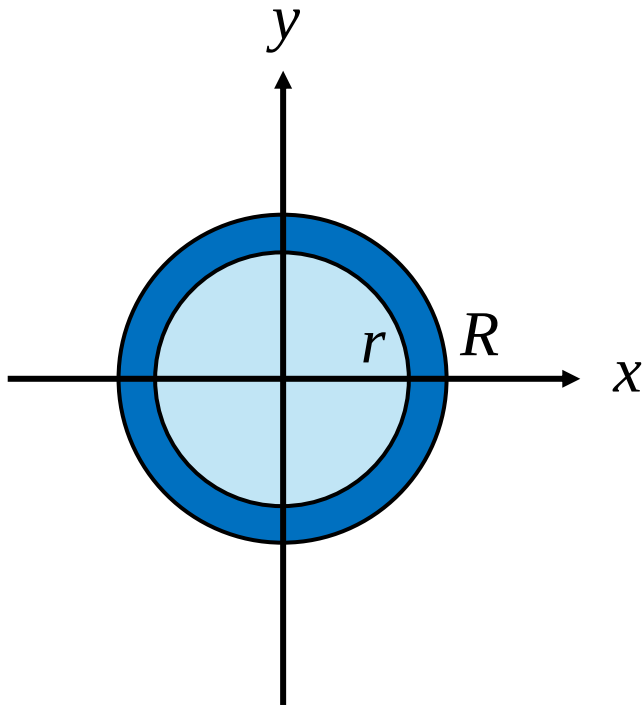


$$\begin{aligned} I_x &= \int_A y^2 dA \\ &= \int_{-\frac{h}{2}}^{\frac{h}{2}} y^2 b dy \\ &= b \left. \frac{y^3}{3} \right|_{-\frac{h}{2}}^{\frac{h}{2}} \\ &= \frac{b}{3} \left(\frac{h^3}{8} - \left(-\frac{h^3}{8} \right) \right) \\ &= \frac{bh^3}{12} \end{aligned}$$

$$\begin{aligned} I_y &= \int_A x^2 dA \\ &= \int_{-\frac{b}{2}}^{\frac{b}{2}} x^2 h dx \\ &= h \left. \frac{x^3}{3} \right|_{-\frac{b}{2}}^{\frac{b}{2}} \\ &= \frac{h}{3} \left(\frac{b^3}{8} - \left(-\frac{b^3}{8} \right) \right) \\ &= \frac{hb^3}{12} \end{aligned}$$

Example

Determine the moment of inertia of a circle/hollow tube



Due to symmetry

$$I_p = I_x + I_y$$

$$I_x = I_y$$

Then

$$I_x = I_y = \frac{I_p}{2} = \begin{cases} \frac{\pi R^4}{4} \\ \frac{\pi}{4} (R^4 - r^4) \end{cases}$$

Parallel axis theorem

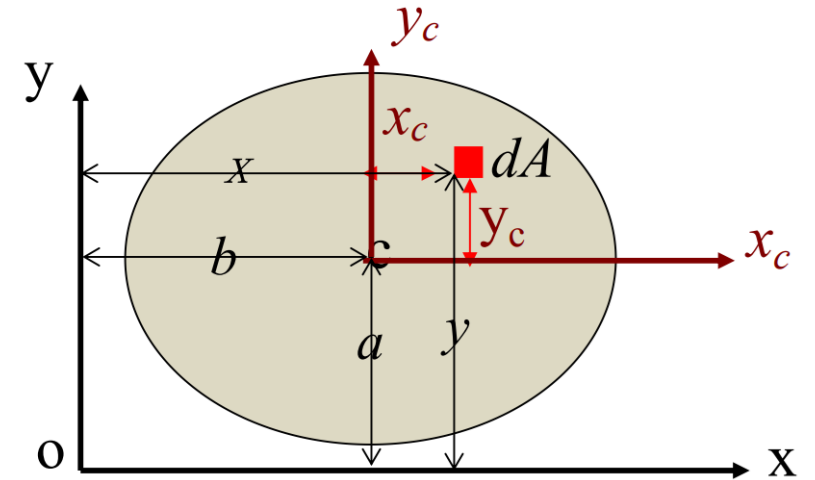
The **parallel axis theorem**, also known as *Huygens–Steiner theorem*, or just as *Steiner's theorem*.

$$I_X = I_{X_c} + Ad^2$$

I_{X_c} : moment of inertia of X_c axis parallel to X axis passing its centroid

A : total area

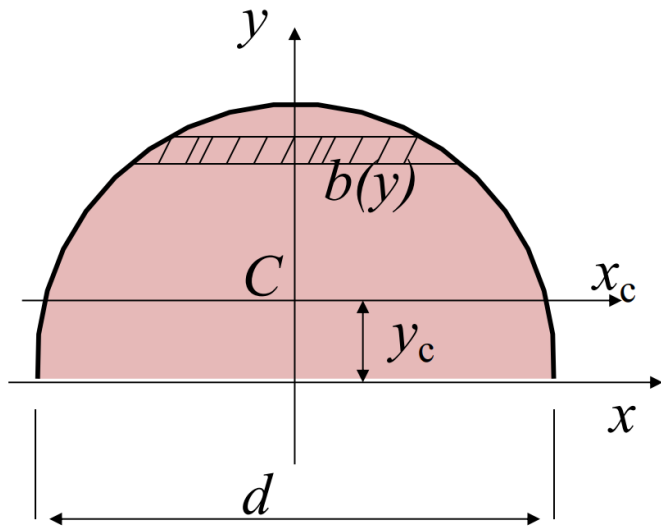
d : distance between two parallel axis X and X_c



Exercise

Determine the moment of inertia of the semicircle with respect to the centroidal axis x_c

$$y_c = \frac{S_x}{A} = \frac{d^3/12}{\pi d^2/8} = \frac{2d}{3\pi}$$



Moment of inertia of a whole circle passing x_c axis

$$I_x = \frac{\pi R^4}{4}$$

Due to symmetry, half circle gives

$$\hat{I}_x = \frac{I_x}{2} = \frac{\pi R^4}{8} = \frac{\pi d^4}{128}$$

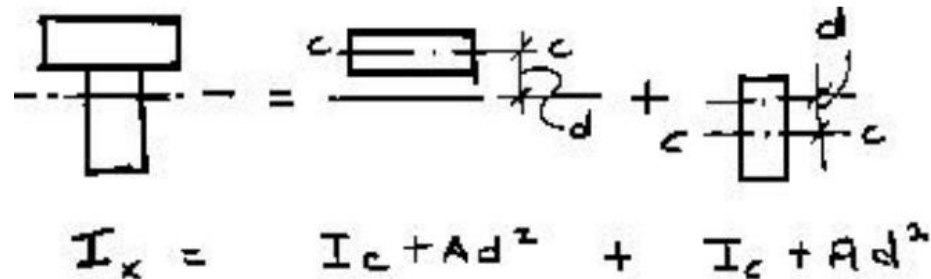
Then

$$\hat{I}_x = \hat{I}_{x_c} + Ay_c^2 \Rightarrow \hat{I}_{x_c} = \hat{I}_x - Ay_c^2 = \frac{\pi d^4}{128} - \frac{d^4}{18\pi}$$

Build-up sections

- It is often advantageous to combine several smaller members to create a beam or column of greater strength
- The moment of inertia of such a built-up section is found by adding the moments of inertia of the component parts

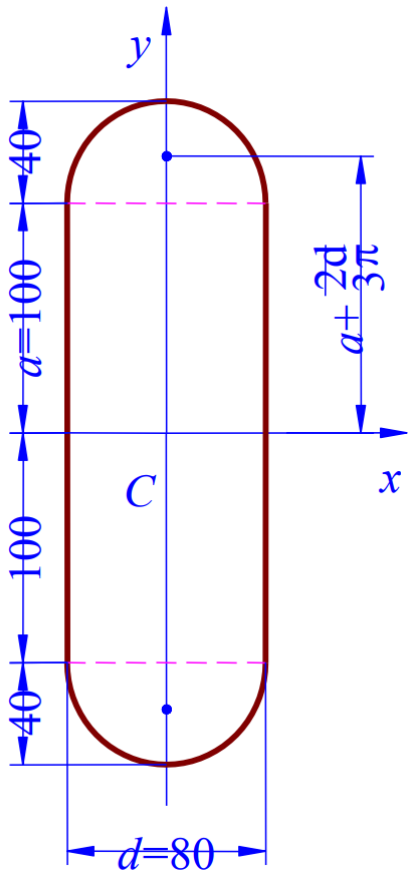
$$I_x = \sum_{i=1}^n I_{xi}, \quad I_y = \sum_{i=1}^n I_{yi}$$



$$I_x = I_c + Ad^2 + I_c + Ad^2$$

Exercise

Determine the inertia with respect to the x axis



Solution: consider the figure as one rectangular and two semicircle.

1) Rectangular's moment of inertia with respect to x axis

$$I_{x1} = \frac{d(2a)^3}{12} = \frac{80 \times 200^3}{12} = 5333 \times 10^4 \text{ mm}^4$$

2) A semicircle's moment of inertia with respect to centroidal axis

$$I_{x_c} = I_x - (y_c)^2 \cdot \frac{\pi d^2}{8} = \frac{\pi d^4}{128} - \frac{d^4}{18\pi}$$

3) A semicircle's moment of inertia with respect to x axis

From Parallel axis theorem:

$$I_{x2} = I_{x_c} + \left(a + \frac{2d}{3\pi}\right)^2 \cdot \frac{\pi d^2}{8} = \frac{\pi d^2}{4} \left(\frac{d^2}{32} + \frac{a^2}{2} + \frac{2ad}{3\pi}\right) = 3467 \times 10^4 \text{ mm}^4$$

4) Whole figure's moment of inertia with respect to x axis

$$I_x = I_{x1} + 2I_{x2} = 5333 \times 10^4 + 2 \times 3467 \times 10^4 = 12270 \times 10^4 \text{ mm}^4$$